1D Arrays Practice Sheet

Easy

1. Input and Output of Array

Take input of n elements in an array and print them.

2. Sum of Array Elements

Calculate the sum of all elements in an array.

3. Find Maximum Element

Find the largest element in the array.

4. Find Minimum Element

Find the smallest element in the array.

5. Search Element

Search for a given element and print its index (or -1 if not found).

6. Count Even and Odd

Count the number of even and odd numbers in the array.

7. Reverse Array

Reverse the elements of the array in-place.

Medium

1. Second Largest Element

Find the second largest element in an array.

2. Sort Array (Bubble Sort)

Sort an array using the bubble sort algorithm.

3. Remove Duplicates

Remove duplicate elements from an array.

4. Frequency of Elements

Print the frequency of each unique element in the array.

5. Left Rotate Array by One

Rotate the array left by one position.

6. Right Rotate Array by K

Rotate the array right by K positions.

7. Check Palindromic Array

Check whether the array is a palindrome (same forward and backward).

Hard

1. Find Unique Element

Given an array where every element appears twice except one, find that unique element.

2. Pair Sum

Find all pairs in the array that sum to a given target value.

3. Subarray with Given Sum

Find a continuous subarray whose sum equals a given number.

4. Maximum Subarray (Kadane's Algorithm)

Find the contiguous subarray with the maximum sum.

5. Trapping Rain Water

Given heights of bars, calculate the amount of trapped rain water.

6. Merge Two Sorted Arrays

Merge two sorted arrays into a single sorted array.

7. Move Zeros to End

Move all zero elements to the end while maintaining the order of non-zero elements.